

1. DEFINITION

DEFINITION
Triglycerol diisostearate EP / Polyglyceryl-3 diisostearate NF

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Polyglyceryl-3 Diisostearate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	46P231IQV8
ADDITIONAL INFORMATION ON THE DENOMINATION	TRIGLYCERYL DIISOSTEARATE ISOCTADECANOIC ACID, DIESTER WITH TRIGLYCEROL
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between Polyglycerol-3 and isostearic acid.

COMPOSITION	
Well defined multi-constituent substance composed of Polyglycerol-3 esters of Isostearic acid (C18iso), the diester fraction being predominant.	
Full Labeling (Key letters system) $A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$	
Polyglyceryl-3 Diisostearate	$A_1 \geq 75.0\%$
Other constituents (i.e. incidental ingredients)	
None	



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4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	11696 - Inactive
For submission of the DMF in another country, please contact your local Gattefossé representative	

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopeia Current edition	Triglycerol diisostearate
USP/ NF Current edition	Polyglyceryl-3 diisostearate

6. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Polyglyceryl-3 diisostearate	Available for use as an Excipient Ingredient in: Devices, Export Only, Listed Medicines, Over the Counter	Only for use in topical medicines for dermal application. Listed Medicines
AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	NHPID name	Route of administration/ Restrictions
Approved chemical name	Polyglyceryl-3 Oleate	Role Non-medicinal: Purposes: Emulsifying Agent , Skin-Conditioning Agent - Emollient General: Route Of Administration: Topical Detail: Topical use only
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		



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7. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE

Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute irritation (Patch test)	Dermal	Human	PLUROL® DIISOSTEARIQUE 10% in stone oil	Well tolerated	Gattefossé data, GAT-92157, 1992
Acute irritation	Ocular	Rabbit	PLUROL® DIISOSTEARIQUE 10% in stone oil	Slightly Irritant	Gattefossé data, GAT-94198, 1994
Sensitisation	Dermal	Human	PLUROL® DIISOSTEARIQUE 10% in stone oil	Not irritant Not Sensitizing	Gattefossé data, GAT-94326, 1994
Acute toxicity (OECD 401)	Oral	Rat	PLUROL® DIISOSTEARIQUE Pure	LD50 _(oral) > 2010 mg/kg	Gattefossé data, GAT-94199, 1994

CIR COMPENDIUM (Cosmetic Ingredient Review)

(<http://www.cir-safety.org>)

Denomination	Reference	Maximum "as used" concentration	Conclusions
Polyglyceryl-3 diisostearate	CIR, 14 Nov 2016	Up to 39 %	The CIR Expert Panel concluded that the 274 polyglyceryl fatty acid esters listed in the assessment are safe in cosmetics in the present practices of use and concentration described in this safety assessment when formulated to be non-irritating.

8. CHEMICAL REGULATIONS

CHEMICAL REGULATION

INCI NAME	CAS #	EINECS #	Chemical name
Polyglyceryl-3 Diisostearate	63705-03-3 (or 66082-42-6)	Not applicable (Not applicable)	1,2,3-Propanetriol, homopolymer, diisooctadecanoate (or Isooctadecanoic acid, diester with triglycerol)

NATIONAL INVENTORIES

INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDL)	Australia (AICS)	China (IECSC)	Other inventories
Polyglyceryl-3 Diisostearate	Listed	DSL listed	Listed	Listed	1,2,3-Propanetriol, homopolymer, diisooctadecanoate (TSCA, DSL, AICS, ECL Serial No. KE-29304, PICCS, ASIA-PAC, NZIoC, IECSC, TCSI) (or Isooctadecanoic acid, diester with triglycerol (AICS, PICCS, ASIA-PAC, NZIoC, TCSI, IECSC))

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting.

Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products



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NATIONAL INVENTORIES					
INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDL)	Australia (AIC)	China (IECSC)	Other inventories
Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.					
Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.					

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
INCI NAME	Status of the product	Status per substance
Polyglyceryl-3 Diisostearate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.

9. HISTORY OF USE

9.1. Pharmaceutical applications

9.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)		
Inactive ingredient	Route / Dosage form	Use level
No FDA IID reference exactly matches the definition polyglyceryl-3 diisostearate		
Maximum potency: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the "maximum potency" field will be blank.		

9.1.2. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.

For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.'

The similarities may be based on the following:

- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);



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- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.

FDA INACTIVE INGREDIENT GUIDE

(<http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm>)

Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
Polyglyceryl-3 oleate (UNII : XHQ165498B)	ORAL	Capsule	NA	2 mg
		Capsule, gelatin coated	NA	1 mg
		Solution	310 mg/mL	NA
	TOPICAL	Cream	2.7 % w/w	NA
	VAGINAL	Cream	2.7 % w/w	NA
Polyglyceryl-4 oleate* (UNII : 15B05TY4GX)	VAGINAL	Emulsion, cream	2.71 %w/w	NA
Polyglyceryl-10 oleate* (UNII: 55C81W76DH)	ORAL	Capsule, soft gelatin	199.9 mg	NA
	ORAL	Solution	190 mg / mL	NA

Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.

**removed from IIG in July 2019 due to new data standard. The reference was not removed from IIG due to safety concerns.*

Liability clause

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